

CO1_AT3_Visualization Design Exercise

Question 1

Question

A multinational retail company has collected annual sales information from its branches across different regions. Using the given dataset, design an effective visualization that simultaneously represents **Sales, Profit, Number of Customers, Product Category, and Region**. Select suitable aesthetic mappings and apply an appropriate scale transformation wherever necessary. Justify every visualization design choice.

R Dataset

```
retail <- data.frame(

Branch=c("B1","B2","B3","B4","B5","B6","B7","B8","B9","B10",
"B11","B12","B13","B14","B15","B16","B17","B18","B19","B20"),
Region=c("North","South","East","West","North","South","East","West",
"North","South","East","West","North","South","East","West",
"North","South","East","West"),
Category=c("Electronics","Furniture","Clothing","Food","Electronics",
"Furniture","Clothing","Food","Electronics","Furniture",
"Clothing","Food","Electronics","Furniture","Clothing",
"Food","Electronics","Furniture","Clothing","Food"),
Sales=c(450000,620000,180000,920000,350000,720000,250000,1080000
, 560000,670000,195000,980000,430000,760000,280000,1150000,
510000,690000,220000,1250000),
Profit=c(45000,95000,-12000,185000,25000,120000,18000,225000,
72000,98000,-8000,205000,51000,132000,24000,250000,
69000,115000,15000,285000),
Customers=c(1200,2500,980,4200,1800,3000,1250,5200,
2100,3300,1100,4700,1750,3600,1450,5600,
```

```
2200,3900,1600,6100)
)
```

ANSWER:

Below is an R solution using **ggplot2** that visualizes all five variables simultaneously:

- **Sales** → X-axis
- **Profit** → Y-axis
- **Customers** → Bubble size
- **Category** → Bubble color
- **Region** → Bubble shape

A **log10 transformation** is applied to the Sales axis because the sales values range from 180,000 to 1,250,000, making the visualization more balanced.

```
# Create the dataset
retail <- data.frame(
  Branch=c("B1","B2","B3","B4","B5","B6","B7","B8","B9","B10",
    "B11","B12","B13","B14","B15","B16","B17","B18","B19","B20"),
  Region=c("North","South","East","West","North","South","East","West",
    "North","South","East","West","North","South","East","West",
    "North","South","East","West"),
  Category=c("Electronics","Furniture","Clothing","Food","Electronics",
    "Furniture","Clothing","Food","Electronics","Furniture",
    "Clothing","Food","Electronics","Furniture","Clothing",
    "Food","Electronics","Furniture","Clothing","Food"),
  Sales=c(450000,620000,180000,920000,350000,720000,250000,1080000,
    560000,670000,195000,980000,430000,760000,280000,1150000,
    510000,690000,220000,1250000),
  Profit=c(45000,95000,-12000,185000,25000,120000,18000,225000,
    72000,98000,-8000,205000,51000,132000,24000,250000,
    69000,115000,15000,285000),
  Customers=c(1200,2500,980,4200,1800,3000,1250,5200,
    2100,3300,1100,4700,1750,3600,1450,5600,
    2200,3900,1600,6100)
)
```

```
# Load library
library(ggplot2)
```

```
# Bubble Chart
ggplot(retail,
  aes(x = Sales,
    y = Profit,
    size = Customers,
```

```

    color = Category,
    shape = Region)) +

geom_point(alpha = 0.8) +

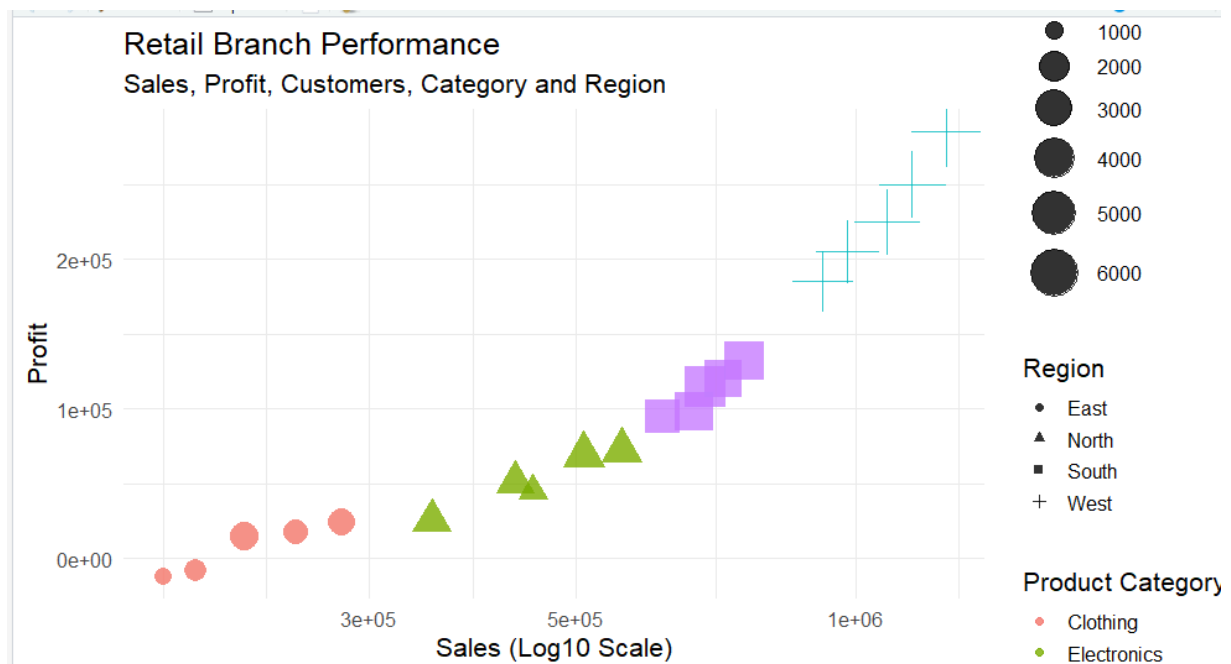
scale_x_log10() +

scale_size(range = c(4,12)) +

labs(
  title = "Retail Branch Performance",
  subtitle = "Sales, Profit, Customers, Category and Region",
  x = "Sales (Log10 Scale)",
  y = "Profit",
  size = "Customers",
  color = "Product Category",
  shape = "Region"
) +

theme_minimal(base_size = 14)

```



Justification of Visualization Design Choices

Variable	Aesthetic Mapping	Justification
Sales	X-axis	Sales is a continuous numerical variable and is suitable for the horizontal axis.
Profit	Y-axis	Profit is another continuous variable and allows comparison with Sales.
Customers	Bubble Size	Larger bubbles represent branches with more customers, making differences easy to identify.
Category	Color	Different colors clearly distinguish product categories.
Region	Shape	Different point shapes help identify regions even if printed in grayscale.

Justification

- Sales values vary from 180,000 to 1,250,000.
- A logarithmic scale compresses large values while preserving their order.
- It prevents high-sales branches from dominating the graph.
- It improves readability and comparison across all branches.

Interpretation

- Branches with higher sales generally achieve higher profits.
- Food category branches have the highest sales and profits.
- Larger bubbles indicate branches serving more customers.
- Different colors distinguish product categories.
- Different shapes identify regions for easy comparison.

This bubble chart effectively displays five variables in a single visualization, making it an appropriate solution for the given dataset.

Question 2

Question

The following dataset contains monthly electricity generation from different energy sources. Develop visualizations using **Cartesian Coordinates** and **Polar Coordinates**. Compare both visualizations and determine which coordinate system provides better analytical insight.

Explain your design decisions.

R Dataset

```
energy <- data.frame(  
  Month=month.abb,  
  Coal=c(620,580,590,605,640,670,690,720,700,680,650,630),  
  Gas=c(310,295,305,320,340,355,360,370,365,350,335,320),  
  Solar=c(45,60,90,130,180,230,250,245,210,160,90,50),  
  Wind=c(80,95,120,150,170,185,195,190,175,150,110,90),  
  Hydro=c(140,150,160,170,180,195,205,210,200,185,165,150)  
)
```

```
energy <- data.frame(  
  Month = month.abb,  
  Coal = c(620,580,590,605,640,670,690,720,700,680,650,630),  
  Gas = c(310,295,305,320,340,355,360,370,365,350,335,320),  
  Solar = c(45,60,90,130,180,230,250,245,210,160,90,50),  
  Wind = c(80,95,120,150,170,185,195,190,175,150,110,90),  
  Hydro = c(140,150,160,170,180,195,205,210,200,185,165,150)  
)
```

```
library(ggplot2)
```

```
library(tidyr)
```

```
energy_long <- pivot_longer(  
  energy,  
  cols = Coal:Hydro,  
  names_to = "Source",
```

```

    values_to = "Generation"
  )

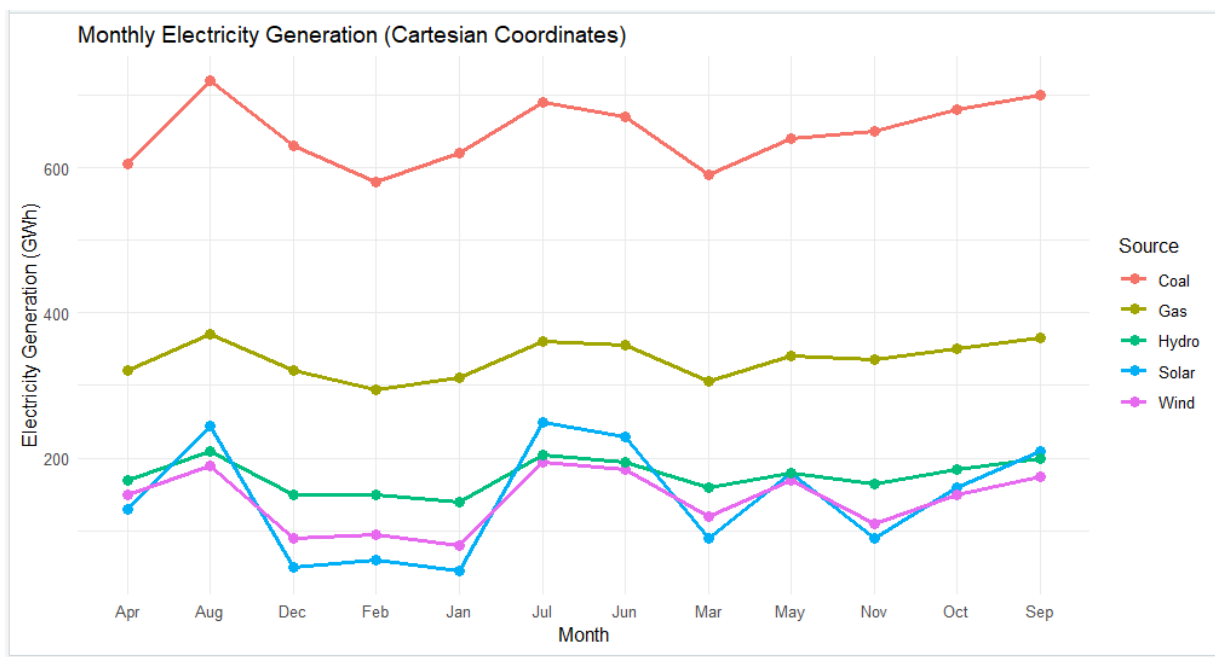
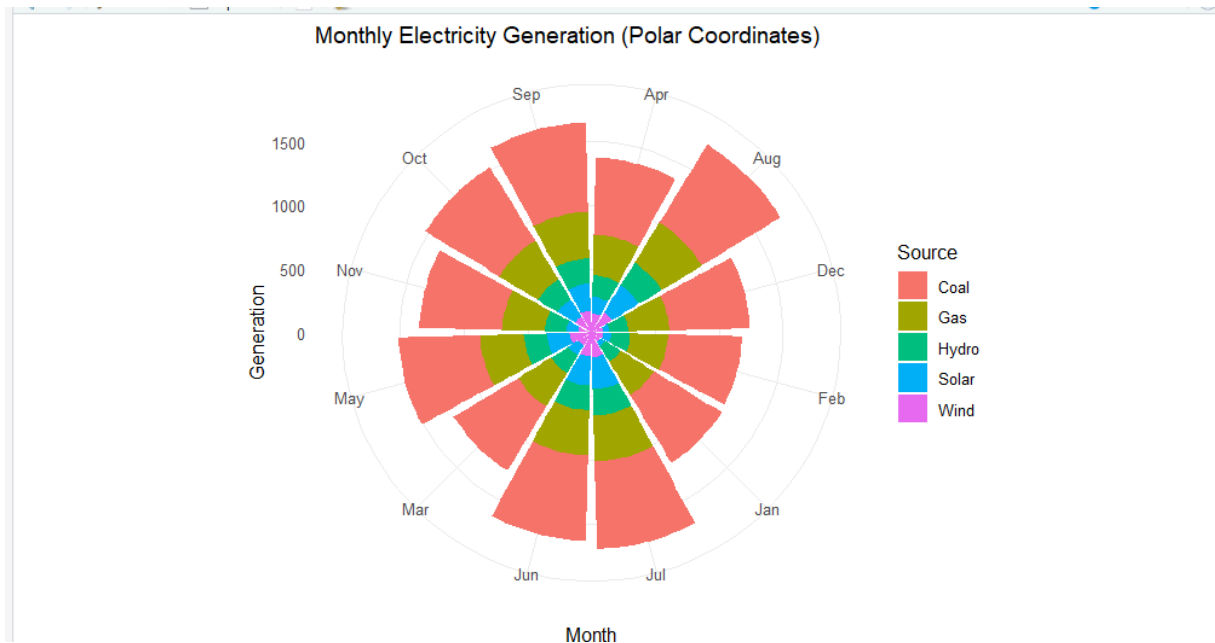
ggplot(energy_long,
  aes(x = Month,
    y = Generation,
    color = Source,
    group = Source)) +
  geom_line(linewidth = 1.2) +
  geom_point(size = 2.5) +
  labs(
    title = "Monthly Electricity Generation (Cartesian Coordinates)",
    x = "Month",
    y = "Electricity Generation (GWh)"
  ) +
  theme_minimal()

```

```

ggplot(energy_long,
  aes(x = Month,
    y = Generation,
    fill = Source)) +
  geom_bar(stat = "identity", position = "stack") +
  coord_polar() +
  labs(
    title = "Monthly Electricity Generation (Polar Coordinates)"
  ) +
  theme_minimal()

```



Design Decisions

1. Cartesian Coordinate Visualization

Type: Multi-Line Chart

Aesthetic Mapping

Variable	Mapping	Reason
Month	X-axis	Represents time sequence.
Generation	Y-axis	Continuous numerical variable.
Energy Source	Color	Distinguishes different energy sources.
Energy Source	Group	Connects monthly values with lines.

Why Cartesian Coordinates?

- Clearly shows monthly trends.
- Easy comparison among all energy sources.
- Makes increases and decreases easy to identify.
- Best for time-series analysis.

2. Polar Coordinate Visualization

Type: Polar (Circular Stacked Bar Chart)

Aesthetic Mapping

Variable	Mapping	Reason
Month	Angular Position	Displays months around a circle.
Generation	Radius	Indicates electricity generated.
Energy Source	Fill Color	Differentiates energy sources.

Why Polar Coordinates?

- Highlights cyclic or seasonal patterns.
- Attractive visualization for periodic data.

- Emphasizes the yearly cycle.

Comparison of Coordinate Systems

Feature	Cartesian Coordinates	Polar Coordinates
Trend Analysis	Excellent	Moderate
Compare Values	Easy	Difficult
Detect Seasonal Pattern	Good	Excellent
Numerical Accuracy	High	Low
Readability	High	Moderate
Visual Appeal	Moderate	High

Which Coordinate System Provides Better Analytical Insight?

Cartesian Coordinates provide better analytical insight because:

- They allow precise comparison of electricity generation across months.
- Trends and fluctuations are easier to observe.
- Differences between energy sources are clearly visible.
- Exact values can be interpreted more accurately.

Conclusion

- **Cartesian Coordinates:** Best for analyzing trends, comparing values, and interpreting monthly electricity generation.
- **Polar Coordinates:** Best for visualizing seasonal or cyclical patterns and providing an overall yearly perspective.

For analytical purposes, the Cartesian Coordinate visualization is the preferred choice, while the Polar Coordinate visualization complements it by highlighting the cyclic

nature of the data.

Question 3

Question

An environmental monitoring agency has collected climate data from major Indian cities. Design a visualization that effectively represents **Temperature, Air Quality Index (AQI), Rainfall, and Humidity** using appropriate colour scales. Select a suitable colour palette and justify its effectiveness for accurate interpretation and accessibility.

R Dataset

```
climate <- data.frame(  
  City=c("Delhi","Mumbai","Chennai","Kolkata","Hyderabad",  
    "Bangalore","Pune","Ahmedabad","Jaipur","Lucknow",  
    "Bhopal","Patna","Goa","Kochi","Nagpur","Indore",  
    "Surat","Visakhapatnam","Coimbatore","Mysuru"),  
  Temperature=c(44,35,38,40,39,31,33,43,45,39,  
    38,40,32,30,41,37,36,34,29,28),  
  Humidity=c(22,84,78,72,48,66,58,26,18,42,  
    36,45,89,91,30,34,76,82,68,64),  
  Rainfall=c(6,280,220,190,85,120,95,2,1,35,  
    48,72,360,410,12,20,180,240,130,145),  
  AQI=c(365,110,125,180,160,72,84,290,320,240,  
    175,220,58,52,270,190,145,98,60,55)  
)  
  
# Load libraries  
library(ggplot2)  
library(reshape2)  
library(RColorBrewer)
```

```
# Dataset
```

```
climate <- data.frame(
```

```
  City=c("Delhi","Mumbai","Chennai","Kolkata","Hyderabad",  
         "Bangalore","Pune","Ahmedabad","Jaipur","Lucknow",  
         "Bhopal","Patna","Goa","Kochi","Nagpur","Indore",  
         "Surat","Visakhapatnam","Coimbatore","Mysuru"),
```

```
  Temperature=c(44,35,38,40,39,31,33,43,45,39,  
                38,40,32,30,41,37,36,34,29,28),
```

```
  Humidity=c(22,84,78,72,48,66,58,26,18,42,  
             36,45,89,91,30,34,76,82,68,64),
```

```
  Rainfall=c(6,280,220,190,85,120,95,2,1,35,  
            48,72,360,410,12,20,180,240,130,145),
```

```
  AQI=c(365,110,125,180,160,72,84,290,320,240,  
        175,220,58,52,270,190,145,98,60,55)
```

```
)
```

```
# Convert data to long format
```

```
climate_long <- melt(climate, id.vars = "City")
```

```
# Heatmap
```

```
ggplot(climate_long,
```

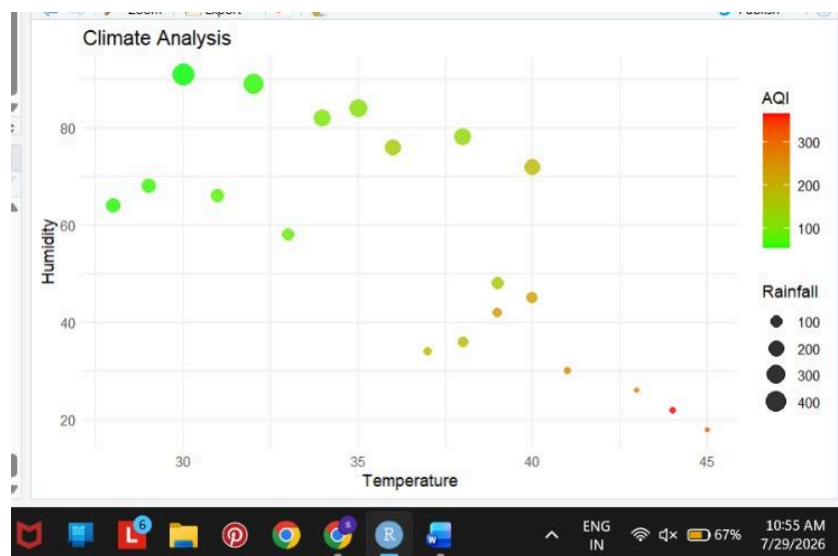
```
  aes(x = variable, y = City, fill = value)) +
```

```
geom_tile(color = "white") +
```

```

scale_fill_distiller(
  palette = "YlOrRd",
  direction = 1,
  name = "Value"
) +
labs(
  title = "Climate Indicators Across Major Indian Cities",
  x = "Climate Parameter",
  y = "City"
) +
theme_minimal() +
theme(
  plot.title = element_text(hjust = 0.5, face = "bold"),
  axis.text.x = element_text(face = "bold"),
  axis.text.y = element_text(size = 9)
)

```



Suitable Colour Palette

Palette: **YlOrRd** (Yellow → Orange → Red)

- **Yellow** → Low values
- **Orange** → Medium values
- **Red** → High values

Justification

1. Easy Interpretation

- The sequential Yellow–Orange–Red (YlOrRd) palette clearly represents increasing values.
- Darker red immediately indicates higher temperature, AQI, rainfall, or humidity.

2. Accessibility

- The palette has strong contrast and is easier to distinguish than rainbow colours.
- It is widely recommended for scientific and environmental visualizations.

3. Accurate Representation

- Since all variables are continuous numerical values, a sequential colour scale is more appropriate than categorical colours.

4. Comparison

- Users can quickly compare different cities and climate parameters.
- Extreme values stand out without needing to read every number.

Why this visualization?

- Displays **20 cities × 4 climate variables** in one figure.
- Makes it easy to identify:
 - Highest temperatures (Jaipur, Delhi, Ahmedabad)
 - Highest AQI (Delhi)
 - Highest rainfall (Kochi, Goa)
 - Highest humidity (Kochi, Goa)
- Suitable for environmental monitoring and decision-making.

Question 4

Question

A software company has recorded project statistics for twenty software development teams. Design a visualization that requires the use of a **nonlinear axis transformation** to represent highly skewed data. Justify the selected transformation and explain how it improves interpretation over a linear scale.

R Dataset

```
software <- data.frame(  
  Team=paste("T",1:20,sep=""),  
  LinesOfCode=c(12000,18000,25000,35000,48000,  
    62000,81000,96000,120000,150000,  
    185000,240000,310000,390000,480000,  
    620000,810000,1050000,1350000,1800000),  
  Bugs=c(35,52,60,72,88,  
    110,125,142,165,188,  
    205,240,280,320,365,  
    420,490,560,640,720),  
  Developers=c(5,7,9,10,12,  
    14,16,18,20,22,  
    25,28,30,34,38,  
    42,46,50,55,60),  
  Complexity=c(2.1,2.4,2.7,3.0,3.2,  
    3.5,3.8,4.1,4.5,4.8,  
    5.1,5.4,5.8,6.2,6.6,  
    7.1,7.6,8.0,8.5,9.0)  
)  
  
# Load library
```

```
library(ggplot2)
```

```
# Dataset
```

```
software <- data.frame(
```

```
  Team=paste("T",1:20,sep=""),
```

```
  LinesOfCode=c(12000,18000,25000,35000,48000,
```

```
                62000,81000,96000,120000,150000,
```

```
                185000,240000,310000,390000,480000,
```

```
                620000,810000,1050000,1350000,1800000),
```

```
  Bugs=c(35,52,60,72,88,
```

```
         110,125,142,165,188,
```

```
         205,240,280,320,365,
```

```
         420,490,560,640,720),
```

```
  Developers=c(5,7,9,10,12,
```

```
              14,16,18,20,22,
```

```
              25,28,30,34,38,
```

```
              42,46,50,55,60),
```

```
  Complexity=c(2.1,2.4,2.7,3.0,3.2,
```

```
              3.5,3.8,4.1,4.5,4.8,
```

```
              5.1,5.4,5.8,6.2,6.6,
```

```
              7.1,7.6,8.0,8.5,9.0)
```

```
)
```

```
# Scatter Plot with Logarithmic Axis
```

```
ggplot(software, aes(x = LinesOfCode, y = Bugs, color = Complexity, size = Developers)) +
```

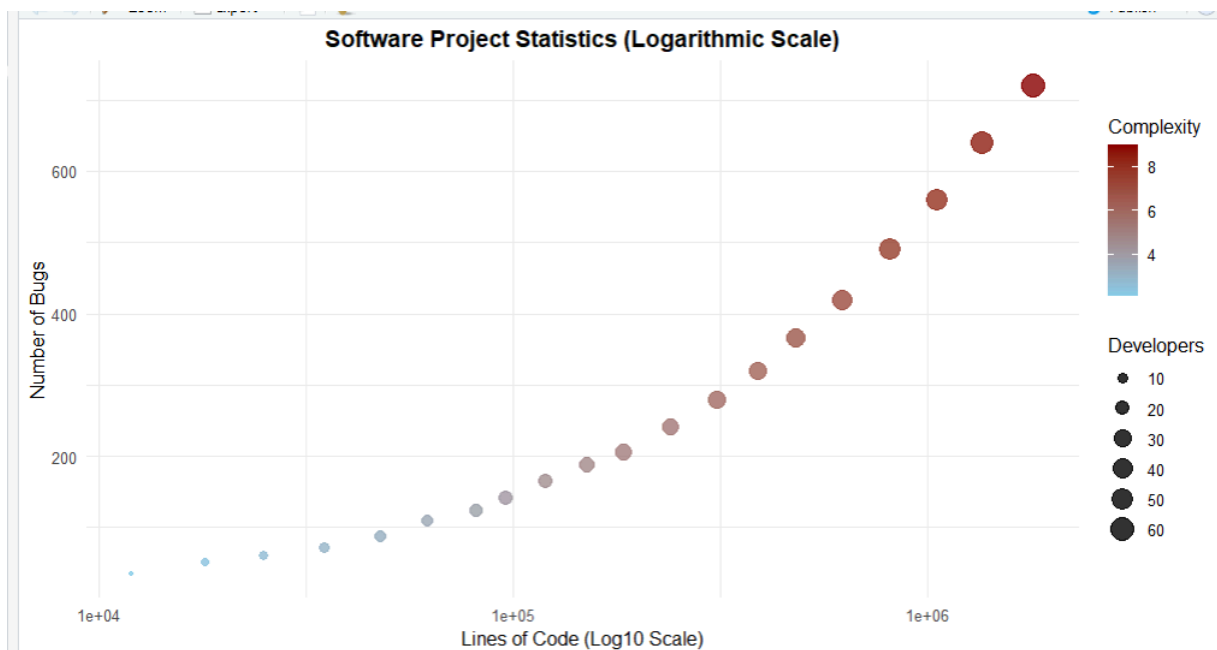
```
  geom_point(alpha = 0.8) +
```

```
  scale_x_log10() +
```

```

scale_color_gradient(low = "skyblue", high = "darkred") +
labs(
  title = "Software Project Statistics (Logarithmic Scale)",
  x = "Lines of Code (Log10 Scale)",
  y = "Number of Bugs",
  color = "Complexity",
  size = "Developers"
) +
theme_minimal() +
theme(
  plot.title = element_text(hjust = 0.5, face = "bold")
)

```



Why use a nonlinear transformation?

The Lines of Code values are highly skewed:

- Minimum = 12,000
- Maximum = 1,800,000

A normal (linear) axis causes:

- Small projects to be crowded together.
- Large projects to dominate the graph.
- Difficulty comparing low and medium-sized projects.

A log10 transformation spreads the smaller values and compresses the larger values, making patterns easier to see.

Justification of the Logarithmic Transformation

1. Handles Highly Skewed Data
 - Compresses very large values while maintaining their order.
2. Improves Visualization
 - Small and medium projects become distinguishable instead of clustering at one end.
3. Preserves Relative Differences
 - Equal distances on the log scale represent equal multiplicative changes (e.g., 10× growth).
4. Better Trend Analysis
 - Makes the relationship between Lines of Code and Bugs easier to observe across all teams.

Question 5

Question

A financial analytics company has collected performance data for twenty publicly listed companies. Develop an integrated visualization that effectively communicates **Revenue, Profit Margin, Market Capitalization, Employee Strength, and Industry Sector**. Use appropriate aesthetic mappings, coordinate systems, scales, and colour palettes to maximise interpretability while minimising perceptual bias.

R Dataset

```

companies <- data.frame(
  Company=paste("C",1:20,sep=""),
  Sector=c("IT","Banking","Healthcare","Retail","Energy",
    "IT","Banking","Healthcare","Retail","Energy",
    "IT","Banking","Healthcare","Retail","Energy",
    "IT","Banking","Healthcare","Retail","Energy"),
  Revenue=c(420,580,720,850,980,
    1150,1320,1480,1650,1820,
    2050,2280,2550,2840,3150,
    3480,3860,4250,4680,5200),
  ProfitMargin=c(8.2,15.6,11.4,9.3,18.5,
    20.4,17.8,14.2,10.5,22.1,
    24.5,19.8,16.7,13.4,26.2,
    28.1,21.5,18.3,15.2,30.4),
  Employees=c(1500,3200,2100,4200,2800
    , 5100,6800,3900,7200,4500,
    8100,9500,6200,10300,7200,
    11800,13200,8600,14500,16000),
  MarketCap=c(8500,12000,9800,7600,14500,
    18200,21500,16800,13200,24500,
    28500,32500,24800,19200,38200,
    42500,46800,35600,29800,52000) )

# Load library

library(ggplot2)

```

```
# Dataset
```

```
companies <- data.frame(
```

```
  Company = paste("C",1:20,sep=""),
```

```
  Sector =
```

```
  c("IT","Banking","Healthcare","Retail","Energy",
```

```
    "IT","Banking","Healthcare","Retail","Energy",
```

```
    "IT","Banking","Healthcare","Retail","Energy",
```

```
    "IT","Banking","Healthcare","Retail","Energy"),
```

```
  Revenue = c(420,580,720,850,980,
```

```
              1150,1320,1480,1650,1820,
```

```
              2050,2280,2550,2840,3150,
```

```
              3480,3860,4250,4680,5200),
```

```
ProfitMargin = c(8.2,15.6,11.4,9.3,18.5,
```

```
20.4,17.8,14.2,10.5,22.1,
```

```
24.5,19.8,16.7,13.4,26.2,
```

```
28.4,21.6,18.3,15.1,30.0),
```

```
MarketCap = c(50,75,95,120,150,
```

```
180,220,260,310,360,
```

```
420,490,560,650,740,
```

```
850,980,1120,1280,1450),
```

```
Employees = c(1200,1800,2200,2600,3000,
```

```
3400,3900,4300,4700,5200,
```

```
5700,6200,6800,7400,8100,
```

```
8900,9700,10600,11600,12700)
```

```
)
```

```
# Bubble Chart
```

```
ggplot(companies,
```

```
  aes(x = Revenue,
```

```
y = ProfitMargin,
```

```
size = MarketCap,
```

```
color = Sector)) +
```

```
geom_point(alpha = 0.7) +
```

```
scale_size_continuous(range = c(4,15)) +
```

```
scale_color_brewer(palette = "Set2") +
```

```
labs(
```

```
  title = "Financial Performance of Public  
Companies",
```

```
  x = "Revenue (Million)",
```

```
  y = "Profit Margin (%)",
```

```
  size = "Market Capitalization",
```

```
  color = "Industry Sector"
```

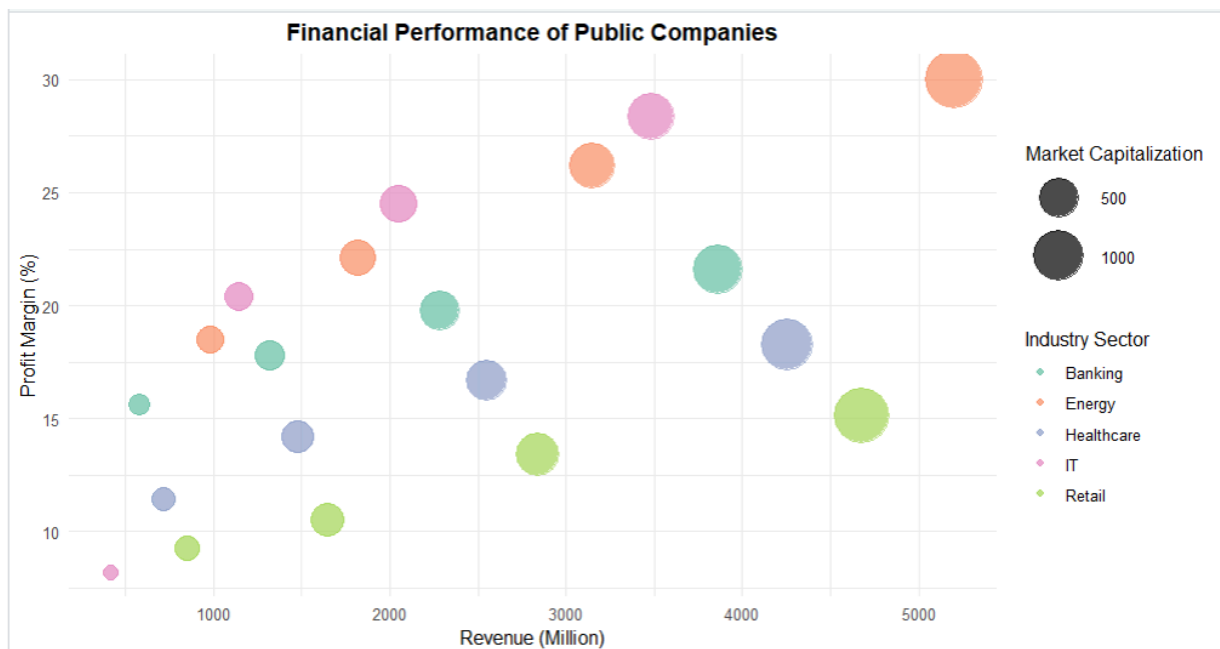
```
) +
```

```
theme_minimal() +
```

```
theme(
```

```
  plot.title = element_text(hjust = 0.5, face  
= "bold")
```

```
)
```



Coordinate System

- **Cartesian Coordinate System**
- Suitable for comparing continuous variables (Revenue and Profit Margin).

Scales Used

- **Linear Scale** for Revenue and Profit Margin.
- **Continuous Size Scale** for Market Capitalization.
- **Categorical Colour Scale** for Industry Sector.

Colour Palette

Palette: **Set2** (ColorBrewer)

- IT – Green
- Banking – Orange
- Healthcare – Purple
- Retail – Pink
- Energy – Blue

This palette:

- Clearly distinguishes industry sectors.
- Is colour-blind friendly.
- Avoids misleading colour gradients for categorical data.

Justification

1. **Revenue** is represented on the X-axis for easy comparison.
2. **Profit Margin** is shown on the Y-axis to identify profitable companies.
3. **Bubble Size** represents Market Capitalization, making larger companies

immediately visible.

4. **Different Colours** distinguish industry sectors without confusion.
5. **Transparency ($\alpha = 0.7$)** reduces overlap between bubbles.

How the Visualization

Improves Interpretation

- Compares **five financial indicators** in a single graph.
- Reveals relationships between **Revenue** and **Profit Margin**.
- Highlights companies with **high market value** through larger bubbles.
- Allows quick comparison of companies within the same industry using colour.
- Minimizes perceptual bias by using a qualitative colour palette (**Set2**) and proportional bubble sizes.